

DBMS Course Project

Department of Computer Science & Engineering

Bangladesh National ID Management System (BNIMS)

11-Week Project Roadmap

A Structured Plan for GitHub-Based Incremental Delivery

Course: Database Management Systems (DBMS)

Project: Bangladesh NID Management System

Duration: 11 Weeks (Weekly GitHub Push)

Tool: Git & GitHub (Version Control)

Date: June 9, 2026

Contents

1	Project Overview	2
1.1	Problem Statement	2
1.2	Objectives	2
2	System Modules	3
2.1	Module Descriptions	3
2.1.1	Citizen Information Module	3
2.1.2	Address Management Module	3
2.1.3	Family Information Module	3
2.1.4	Verification Module	3
2.1.5	Administrative Module	4
3	11-Week Project Roadmap	4
4	Project Phases	5
4.1	Phase 1 — Planning & Requirements (Weeks 1–3)	5
4.2	Phase 2 — Database Design & Implementation (Weeks 4–6)	5
4.3	Phase 3 — Data, Queries & Testing (Weeks 7–9)	6
4.4	Phase 4 — Finalization & Submission (Weeks 10–11)	6
5	GitHub Repository Structure	6
6	Weekly GitHub Push Summary	7
7	Major Deliverables Summary	8
8	Normalization Summary	9
9	Tools & Technologies	9

1. Project Overview

The **Bangladesh National ID Management System (BNIMS)** is a database-driven application designed to manage citizen identity records for Bangladesh. The system covers citizen information, address management, family records, NID verification, and administrative operations. This document outlines the complete 11-week roadmap with weekly deliverables and GitHub push schedule.

1.1. Problem Statement

Bangladesh currently manages National ID (NID) data through fragmented, paper-based, or siloed digital systems that suffer from:

- ▶ Redundant data storage and inconsistent records
- ▶ Lack of a normalized relational database structure
- ▶ Slow verification processes and limited query capability
- ▶ No centralized update history or audit trail
- ▶ Poor scalability for Bangladesh's 170+ million citizens

1.2. Objectives

1. Design a fully normalized relational database (up to 3NF) for NID data.
2. Implement all CRUD operations with referential integrity constraints.
3. Develop a working UI prototype for citizen and administrative use.
4. Provide a comprehensive SQL query library for verification and reporting.
5. Maintain a clean, documented GitHub repository with weekly incremental commits.

2. System Modules

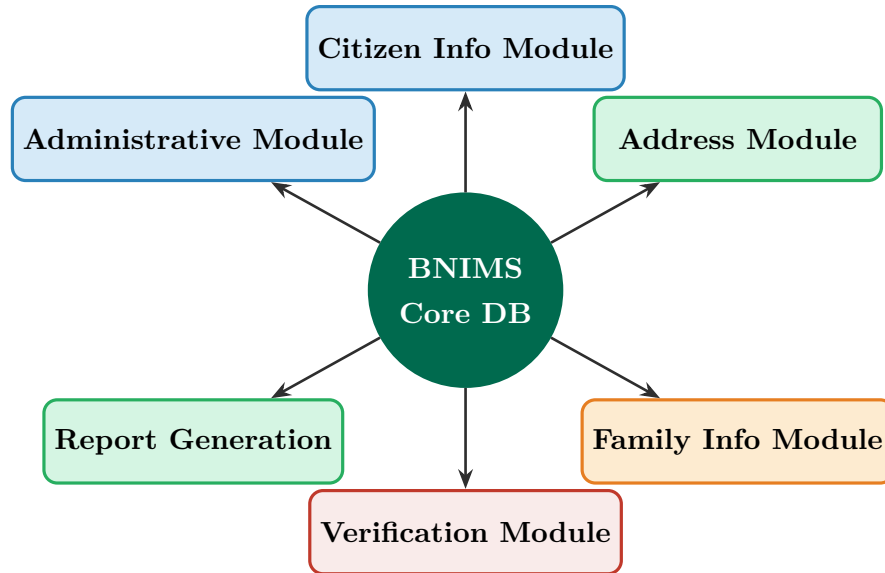


Figure 1: BNIMS Module Architecture

2.1. Module Descriptions

2.1.1. Citizen Information Module

Stores core identity attributes:

- NID Number (Primary Key), Full Name, Date of Birth
- Gender, Blood Group, Marital Status

2.1.2. Address Management Module

Maintains hierarchical address data: Present Address, Permanent Address, Division, District, Upazila.

2.1.3. Family Information Module

Stores relational family references: Father Information, Mother Information, Spouse Information.

2.1.4. Verification Module

Handles NID verification queries and maintains an Information Update History log for audit purposes.

2.1.5. Administrative Module

Provides User Login (role-based access), Data Management operations, and Report Generation capabilities.

3. 11-Week Project Roadmap

Week	Activities	Deliverables / GitHub Push
1	Project Topic Selection, Problem Identification, Requirement Gathering	docs/proposal.md, Problem Statement, Objectives
2	Literature Review and Existing System Analysis	docs/literature_review.md, Comparative Analysis Report
3	Requirement Analysis (Functional & Non-Functional Requirements)	docs/SRS.md — Software Requirement Specification Draft
4	Database Design (ER Diagram, Entity Identification, Relationships)	design/ER_diagram.png, design/data_dictionary.md
5	Relational Schema Design, Normalization (1NF, 2NF, 3NF)	design/schema.md, Normalization Documentation
6	SQL Table Creation, Primary Key & Foreign Key Implementation	sql/create_tables.sql, sql/constraints.sql
7	Data Insertion, Query Development, CRUD Operations	sql/insert_data.sql, sql/queries.sql, Sample Dataset
8	User Interface (UI) Design and Prototype Development	ui/wireframes/, Prototype Screenshots & Mockups
9	System Integration and Testing	tests/test_cases.md, tests/testing_report.md
10	Performance Evaluation, Error Fixing, Documentation Finalization	docs/final_documentation.md, Optimized SQL scripts

Week	Activities	Deliverables / GitHub Push
11	Project Presentation Preparation and Final Submission	presentation/BNIMS_slides.pdf, docs/final_report.pdf

Table 1: BNIMS — 11-Week Weekly Roadmap with GitHub Push Schedule

4. Project Phases

4.1. Phase 1 — Planning & Requirements (Weeks 1–3)

Phase 1: Planning & Requirements

- **Week 1:** Select project topic, define scope, identify stakeholders
- **Week 2:** Review existing NID systems (Bangladesh EC, NADRA, Aadhaar), comparative analysis
- **Week 3:** Finalize SRS with functional and non-functional requirements

GitHub Push: docs/ folder — proposal.md, literature_review.md, SRS.md

4.2. Phase 2 — Database Design & Implementation (Weeks 4–6)

Phase 2: Database Design & Implementation

- **Week 4:** Draw ER diagram with all entities, attributes, and relationships
- **Week 5:** Convert ER to relational schema; apply 1NF, 2NF, 3NF normalization
- **Week 6:** Write DDL SQL scripts; implement PKs, FKs, and constraints

GitHub Push: design/ — ER diagram, schema; sql/ — create_tables.sql

4.3. Phase 3 — Data, Queries & Testing (Weeks 7–9)

Phase 3: Data

- **Week 7:** Insert sample data; develop SELECT, INSERT, UPDATE, DELETE queries
- **Week 8:** Design UI wireframes and prototype screens
- **Week 9:** Integrate modules; write and execute test cases

GitHub Push: `sql/queries.sql`, `ui/wireframes/`, `tests/`

4.4. Phase 4 — Finalization & Submission (Weeks 10–11)

Phase 4: Finalization & Submission

- **Week 10:** Fix identified bugs; optimize queries; finalize all documentation
- **Week 11:** Build presentation slides; complete and submit final report

GitHub Push: `docs/final_documentation.md`,
`presentation/BNIMS_slides.pdf`, `docs/final_report.pdf`

5. GitHub Repository Structure

The project will follow this repository folder structure for weekly pushes:

```

BNIMS/
  README.md
  docs/
    proposal.md
    literature_review.md
    SRS.md
    final_documentation.md
    final_report.pdf
  design/
    ER_diagram.png
    data_dictionary.md
    schema.md
  sql/
    create_tables.sql
    constraints.sql
    insert_data.sql
    queries.sql
  ui/
    wireframes/
  tests/
    test_cases.md
    testing_report.md
  presentation/
  BNIMS_slides.pdf

```

6. Weekly GitHub Push Summary

Week	Files to Push	Phase
1	docs/proposal.md	Phase 1
2	docs/literature_review.md	Phase 1
3	docs/SRS.md	Phase 1
4	design/ER_diagram.png, design/data_dictionary.md	Phase 2
5	design/schema.md	Phase 2
6	sql/create_tables.sql, sql/constraints.sql	Phase 2
7	sql/insert_data.sql, sql/queries.sql	Phase 3

Week	Files to Push	Phase
8	ui/wireframes/	Phase 3
9	tests/test_cases.md, tests/testing_report.md	Phase 3
10	docs/final_documentation.md	Phase 4
11	presentation/BNIMS_slides.pdf, docs/final_report.pdf	Phase 4

Table 2: Weekly GitHub Push Schedule

7. Major Deliverables Summary

Phase	Key Deliverables	Output Format
Phase 1 (Wk 1–3)	Project Proposal, Objectives, SRS Document	Markdown / PDF
Phase 2 (Wk 4–6)	ER Diagram, Relational Schema, Normalization, SQL DDL	PNG, MD, SQL
Phase 3 (Wk 7–9)	CRUD Queries, UI Prototype, Test Report	SQL, PNG, MD
Phase 4 (Wk 10–11)	Final Report, Presentation, Demo	PDF, PPTX

Table 3: Major Deliverables by Phase

8. Normalization Summary

Normal Form	Rule Applied	BNIMS Action
1NF	Atomic values; no repeating groups	Separate multi-valued address/family fields
2NF	No partial dependency on composite key	All non-key attributes depend on full PK
3NF	No transitive dependencies	Extract Division, District, Upazila tables

Table 4: Normalization Steps Applied in BNIMS Database Design

9. Tools & Technologies

Category	Tool / Technology
Database	MySQL / PostgreSQL
ER Diagram	draw.io / dbdiagram.io / Lucidchart
UI Prototype	Figma / Balsamiq
Documentation	Markdown, L ^A T _E X
Version Control	Git & GitHub
SQL Client	MySQL Workbench / pgAdmin / DBeaver
Reporting	PDF (LaTeX / Pandoc)

Table 5: Tools & Technologies Used in BNIMS